

Curvature elasticity and vesicle stability

These notes reconstruct the calculations in *Physical Review A* **39**, 5280–5288 (1989) [1]. Tags (1)–(97) refer to that paper’s equations; lettered tags supply intermediate algebra. The notation, curvature orientation, perturbation order, and physical scope follow the paper. In particular, $\delta^{(2)}F$ and $\delta^{(3)}F$ denote the quadratic and cubic terms in the energy increment, including their Taylor factors. They do not mean the second and third derivatives without those factors.

1 Geometry and variations of area and volume

The energy is

$$F = \frac{k_c}{2} \oint (c_1 + c_2 - c_0)^2 dA + \Delta p \int dV + \lambda \oint dA. \quad (1)$$

Here k_c is the bending rigidity, c_0 the spontaneous curvature, Δp the exterior pressure minus the interior pressure, and λ the area multiplier or imposed tension. The surface is closed, its topology is fixed, and these coefficients are constant. The original calculation does not include a separate Gaussian-curvature energy.

Use coordinates u, v and the embedding $\mathbf{Y}(u, v)$, with repeated indices summed from 1 to 2:

$$\begin{aligned} \mathbf{Y}_i &= \partial_i \mathbf{Y}, & \mathbf{Y}_{ij} &= \partial_i \partial_j \mathbf{Y}, & g_{ij} &= \mathbf{Y}_i \cdot \mathbf{Y}_j, & g &= \det(g_{ij}), \\ g^{ij} &= (g_{ij})^{-1}, & L_{ij} &= \mathbf{Y}_{ij} \cdot \mathbf{n}, & L &= \det(L_{ij}), & L^{ij} &= (L_{ij})^{-1}. \end{aligned} \quad (2)$$

The notation L^{ij} means the matrix inverse of L_{ij} , rather than merely raising its indices with g^{ij} . Products KL^{ij} can be evaluated as cofactors divided by g , including when one principal curvature vanishes. The normal, connection, and curvatures are

$$\mathbf{n} = \frac{\mathbf{Y}_1 \times \mathbf{Y}_2}{\sqrt{g}}, \quad \mathbf{Y}_{ij} = \Gamma_{ij}^k \mathbf{Y}_k + L_{ij} \mathbf{n}, \quad (3)$$

$$H = -\frac{1}{2}(c_1 + c_2) = \frac{1}{2}g^{ij}L_{ij}, \quad K = c_1 c_2 = \frac{L}{g}. \quad (4)$$

Thus $H = -1/r_0$ on an outward-oriented sphere. A normal perturbation is

$$\mathbf{Y}' = \mathbf{Y} + \Psi(u, v)\mathbf{n}. \quad (5)$$

A subscript on Ψ denotes an ordinary coordinate derivative. Differentiating and then using the Weingarten equation gives

$$\mathbf{Y}'_i = \mathbf{Y}_i + \Psi_i \mathbf{n} + \Psi \partial_i \mathbf{n}, \quad (6)$$

$$\partial_i \mathbf{n} = -L_{ij} g^{jk} \mathbf{Y}_k, \quad (7)$$

$$\mathbf{Y}'_i = \mathbf{Y}_i + \Psi_i \mathbf{n} - \Psi L_{ij} g^{jk} \mathbf{Y}_k. \quad (8)$$

The two-dimensional characteristic identity is

$$L_{ij} g^{jk} L_{kl} = 2H L_{il} - K g_{il}. \quad (9)$$

Indeed its eigenvalues obey $c_a^2 - (c_1 + c_2)c_a + c_1c_2 = 0$, with the sign of the eigenvalues of $g^{-1}L$ fixed by (4). Taking the scalar product of (8) with itself now gives, without an approximation,

$$\delta g_{ij} = -2\Psi L_{ij} + \Psi_i \Psi_j + \Psi^2(2H L_{ij} - K g_{ij}). \quad (10)$$

For a two-dimensional determinant,

$$g = \frac{1}{2} e_{3ij} e_{3kl} g_{ik} g_{jl}, \quad (11)$$

$$\delta g = g[-4H\Psi + g^{ij}\Psi_i\Psi_j + (4H^2 + 2K)\Psi^2] + O(\Psi^3). \quad (12)$$

The alternating symbol in (11) is specified by

$$e_{123} = e_{231} = e_{312} = 1, \quad e_{132} = e_{213} = e_{321} = -1, \quad e_{ijk} = 0 \quad (i = j \vee j = k \vee k = i). \quad (13)$$

For example, the linear determinant term is $g^{ij}(-2\Psi L_{ij}) = -4H\Psi$. At quadratic order the trace of the quadratic part of (10) is $g^{ij}\Psi_i\Psi_j + (4H^2 - 2K)\Psi^2$; the determinant of its linear part adds $4K\Psi^2$.

Expanding the matrix inverse using $g^{ik}g'_{kj} = \delta_j^i$ yields

$$\begin{aligned} \delta g^{ij} &= 2\Psi(2H g^{ij} - K L^{ij}) + \left(\frac{e_{3ik}e_{3jl}}{g} - g^{ij}g^{kl} \right) \Psi_k \Psi_l \\ &\quad - 3\Psi^2[(K - 4H^2)g^{ij} + 2HKL^{ij}] + O(\Psi^3). \end{aligned} \quad (14)$$

The underlying inverse identities are

$$\begin{aligned} \delta^{(1)} g^{ij} &= -g^{ik}(\delta^{(1)} g_{kl})g^{lj}, \\ \delta^{(2)} g^{ij} &= -g^{ik}(\delta^{(2)} g_{kl})g^{lj} + g^{ik}(\delta^{(1)} g_{kl})g^{lm}(\delta^{(1)} g_{mn})g^{nj}. \end{aligned} \quad (14a)$$

Expanding the perturbed normal from $\mathbf{n}' \cdot \mathbf{Y}'_i = 0$ and $\mathbf{n}' \cdot \mathbf{n}' = 1$ gives the ingredients for L'_{ij} :

$$\mathbf{n}' = \left(1 - \frac{1}{2}g^{ij}\Psi_i\Psi_j \right) \mathbf{n} - g^{ij}\Psi_i\mathbf{Y}_j - \Psi g^{ij}\Psi_i L_{jk}g^{kl}\mathbf{Y}_l + O(\Psi^3). \quad (15a)$$

Using $L'_{ij} = \mathbf{Y}'_{ij} \cdot \mathbf{n}'$ and collecting terms gives the paper's component expansion,

$$\begin{aligned} \delta L_{ij} &= \Psi_{ij} + \Psi(K g_{ij} - 2H L_{ij}) - \Gamma_{ij}^k \Psi_k \\ &\quad + \Psi \Psi_m [K \Gamma_{ij}^k e_{3lk} e_{3mp} g_{pq} L^{ql} + (L_{jl} g^{lm})_i + L_{jk} g^{kl} \Gamma_{li}^m] \\ &\quad + \Psi_k \Psi_m [g^{lk}(\delta_{im} L_{jl} + \delta_{jm} L_{il}) - \frac{1}{2} L_{ij} g^{mk}] + O(\Psi^3). \end{aligned} \quad (15)$$

Here $(L_{jl} g^{lm})_i = \partial_i (L_{jl} g^{lm})$. Contracting $2H' = g^{ij} L'_{ij}$ requires the linear terms, the quadratic terms, and the product $(\delta^{(1)} g^{ij})(\delta^{(1)} L_{ij})$:

$$\begin{aligned} \delta H &= \Psi(2H^2 - K) + \frac{1}{2}g^{ij}\nabla_i\Psi_j \\ &\quad + \frac{1}{2}\Psi\Psi_m [g^{ij}(L_{jl} g^{lm})_i - L_{lk} g^{km} g^{ij} \Gamma_{ij}^l + (KL^{ij} - 2H g^{ij}) \Gamma_{ij}^m] \\ &\quad + \Psi^2(4H^3 - 3HK) + \frac{1}{2}\Psi_i\Psi_j(H g^{ij} - KL^{ij}) + \Psi\Psi_{ij}(2H g^{ij} - KL^{ij}) + O(\Psi^3), \end{aligned} \quad (16)$$

where

$$\nabla_i \Psi_j = \Psi_{ij} - \Gamma_{ij}^k \Psi_k. \quad (17)$$

For checking the contraction, its linear part is $\frac{1}{2}[4H\Psi H - 2K\Psi + g^{ij}\nabla_i\Psi_j]$. The square-root expansion of (12) gives

$$\delta\sqrt{g} = \sqrt{g} \left[-2H\Psi + \frac{1}{2}g^{ij}\Psi_i\Psi_j + K\Psi^2 \right] + O(\Psi^3), \quad (18)$$

$$\delta A = \oint \left[-2H\Psi + \frac{1}{2}g^{ij}\Psi_i\Psi_j + K\Psi^2 \right] dA + O(\Psi^3). \quad (19)$$

In particular, the $2H^2\Psi^2$ term from half of (12) cancels the $2H^2\Psi^2$ from the square of its linear part. The volume swept out by a normal segment gives

$$\delta V = \oint (\Psi - H\Psi^2) dA + O(\Psi^3). \quad (20)$$

Its integrand is the integral from 0 to Ψ of the parallel-surface area factor $1 - 2H\Psi + K\Psi^2$, with the integration variable understood as the normal distance along that segment.

2 The shape equation and the general quadratic energy

With the sign convention (4), the curvature energy is $F_c = \frac{1}{2}k_c \oint (2H + c_0)^2 dA$. Therefore

$$\delta^{(1)}F = \Delta p \delta^{(1)}V + \lambda \delta^{(1)}A + \frac{k_c}{2} \delta^{(1)} \oint (2H + c_0)^2 dA. \quad (21)$$

Equations (19) and (20) immediately give

$$\delta^{(1)}V = \oint \Psi dA, \quad (22)$$

$$\delta^{(1)}A = - \oint 2H\Psi dA. \quad (23)$$

The product rule for the energy density and area element gives

$$\delta^{(1)}F_c = \frac{k_c}{2} \oint [(2H + c_0)^2 \delta^{(1)}dA + 4(2H + c_0)(\delta^{(1)}H)dA]. \quad (24)$$

From (16),

$$\delta^{(1)}H = \Psi(2H^2 - K) + \frac{1}{2}g^{ij}\nabla_i\Psi_j, \quad (25)$$

$$\delta^{(1)}H = \Psi(2H^2 - K) + \frac{1}{2}g^{ij}(\Psi_{ij} - \Gamma_{ij}^k \Psi_k). \quad (26)$$

The nondifferentiated part of (24) simplifies as follows:

$$\begin{aligned} & -k_c H(2H + c_0)^2 + 2k_c(2H + c_0)(2H^2 - K) \\ & = k_c(2H + c_0)(2H^2 - 2K - c_0H). \end{aligned} \quad (26a)$$

Twice integrating Ψ_{ij} by parts and once integrating Ψ_k gives

$$\begin{aligned} \delta^{(1)}F &= \oint \Psi \left[\Delta p - 2\lambda H + k_c(2H + c_0)(2H^2 - 2K - c_0H) \right. \\ &\quad \left. + \frac{k_c}{\sqrt{g}}(\partial_i \partial_j + \partial_k \Gamma_{ij}^k)[g^{ij}\sqrt{g}(2H + c_0)] \right] dA. \end{aligned} \quad (27)$$

In (27), $\partial_k \Gamma_{ij}^k$ is an operator composition: ∂_k acts on Γ_{ij}^k and on everything following it. Metric compatibility supplies the identity used in the paper,

$$\partial_i \{ [\partial_j (g^{ij} \sqrt{g})] f \} = -\partial_k [\Gamma_{ij}^k g^{ij} \sqrt{g} f], \quad (28)$$

The remaining differential term is consequently $k_c \Delta (2H + c_0) = 2k_c \Delta H$. Thus

$$\delta^{(1)} F = \oint \Psi [\Delta p - 2\lambda H + k_c (2H + c_0) (2H^2 - 2K - c_0 H) + 2k_c \Delta H] dA, \quad (29)$$

$$\Delta = \frac{1}{\sqrt{g}} \partial_i (g^{ij} \sqrt{g} \partial_j). \quad (30)$$

The same integration by parts can be checked directly as

$$\oint (2H + c_0) \Delta \Psi dA = - \oint \nabla (2H + c_0) \cdot \nabla \Psi dA = 2 \oint \Psi \Delta H dA. \quad (30a)$$

Arbitrary Ψ then gives

$$\Delta p - 2\lambda H + k_c (2H + c_0) (2H^2 - 2K - c_0 H) + 2k_c \Delta H = 0. \quad (31)$$

Collecting the quadratic terms. Keep the multipliers fixed at their equilibrium values while expanding:

$$\delta^{(2)} F = \Delta p \delta^{(2)} V + \lambda \delta^{(2)} A + \frac{k_c}{2} \delta^{(2)} \oint (2H + c_0)^2 dA, \quad (32)$$

$$\delta^{(2)} V = - \oint H \Psi^2 dA, \quad (33)$$

$$\delta^{(2)} A = \oint (\frac{1}{2} g^{ij} \Psi_i \Psi_j + K \Psi^2) dA. \quad (34)$$

There are four terms in the curvature-energy product rule:

$$\begin{aligned} \delta^{(2)} F_c &= \frac{k_c}{2} \oint [4(2H + c_0) \delta^{(2)} H + 4(\delta^{(1)} H)^2] dA \\ &+ \frac{k_c}{2} \oint [(2H + c_0)^2 \delta^{(2)} dA + 4(2H + c_0) (\delta^{(1)} H) \delta^{(1)} dA], \end{aligned} \quad (35)$$

$$\delta^{(2)} dA = (\frac{1}{2} g^{ij} \Psi_i \Psi_j + K \Psi^2) dA. \quad (36)$$

To eliminate mixed factors with undifferentiated Ψ , use

$$\Psi \Psi_i = \frac{1}{2} \partial_i \Psi^2, \quad (37)$$

$$\Psi \Psi_{ij} = \frac{1}{2} \partial_i \partial_j \Psi^2 - \Psi_i \Psi_j. \quad (38)$$

For instance, the second identity changes a $\Psi \Psi_{ij}$ term into a contribution to Ψ^2 after two integrations by parts and a contribution to $\Psi_i \Psi_j$ with a minus sign. Substitution of (16),

(25), (33), and (36) into (35), followed by exactly these integrations, gives the full expression

$$\begin{aligned}
\delta^{(2)}F = \oint \Big\{ & \Psi^2 \Big[\lambda K - \Delta p H + 2k_c(H + \tfrac{1}{2}c_0)(8H^3 - 5KH + \tfrac{1}{2}c_0K) \\
& + 2k_c(K - 2H^2)(K + 2c_0H + 2H^2) \\
& + \frac{2k_c}{\sqrt{g}}\partial_i\partial_j[\sqrt{g}(H + \tfrac{1}{2}c_0)(2Hg^{ij} - KL^{ij})] \\
& - \frac{k_c}{\sqrt{g}}\partial_m \Big\{ \sqrt{g}(H + \tfrac{1}{2}c_0)[g^{ij}\partial_i(L_{jl}g^{lm}) \\
& \quad - L_{lk}g^{km}g^{ij}\Gamma_{ij}^l - (2Hg^{ij} - KL^{ij})\Gamma_{ij}^m] \Big\} \\
& + \Psi_i\Psi_j \Big[(\tfrac{1}{2}\lambda + k_c(H + \tfrac{1}{2}c_0)^2)g^{ij} + 2k_c(H + \tfrac{1}{2}c_0)(KL^{ij} - 3Hg^{ij}) \Big] \\
& - 2k_c(K + c_0H)\Psi g^{ij}\nabla_i\Psi_j + \frac{k_c}{2}(g^{ij}\nabla_i\Psi_j)^2 \Big\} dA.
\end{aligned} \tag{39}$$

Every term is quadratic in Ψ . In particular, $4(\delta^{(1)}H)^2$ produces the last square, while its cross term and the first-order area correction contribute to the $\Psi\Delta\Psi$ term. The next two sections evaluate this expression on precisely the two background geometries treated in the original paper.

3 Sphere: mode energies and the pressure threshold

For a sphere, $H = -1/r_0$, $K = 1/r_0^2$, and $\Delta H = 0$. Substituting in (31) and multiplying by r_0^3 gives

$$\Delta p r_0^3 + 2\lambda r_0^2 - k_c c_0 r_0(2 - c_0 r_0) = 0. \tag{40}$$

The spherical embedding is

$$\mathbf{Y} = r_0(\cos\phi\sin\theta, \sin\phi\sin\theta, \cos\theta), \tag{41}$$

with $u = \theta$, $v = \phi$. Direct differentiation gives

$$\begin{aligned}
g_{11} &= r_0^2, & g_{12} &= g_{21} = 0, & g_{22} &= r_0^2 \sin^2\theta, \\
L_{11} &= -r_0, & L_{12} &= L_{21} = 0, & L_{22} &= -r_0 \sin^2\theta, \\
\Gamma_{11}^1 &= \Gamma_{11}^2 = \Gamma_{12}^1 = \Gamma_{22}^2 = 0, & \Gamma_{12}^2 &= \cot\theta, & \Gamma_{22}^1 &= -\sin\theta\cos\theta, \\
H &= -r_0^{-1}, & K &= r_0^{-2}.
\end{aligned} \tag{42}$$

Hence

$$\Delta_s = \frac{1}{r_0^2 \sin\theta} \partial_\theta(\sin\theta \partial_\theta) + \frac{1}{r_0^2 \sin^2\theta} \partial_\phi^2, \tag{43}$$

$$\Delta_s \Psi = g^{ij}\nabla_i\Psi_j = -r_0^{-1}L^{ij}\nabla_i\Psi_j. \tag{44}$$

These substitutions turn (39) into

$$\delta^{(2)}F = \oint \Psi [D + B\Delta_s + \tfrac{1}{2}k_c\Delta_s^2] \Psi dA, \tag{45}$$

$$D = \frac{\Delta p}{2r_0} + \frac{k_c c_0}{r_0^3}, \quad B = \frac{\Delta p r_0}{4} + \frac{k_c(2 + c_0 r_0)}{2r_0^2}. \quad (46)$$

Here (40) was used to eliminate λ . An equivalent way to check the integration by parts is

$$\delta^{(2)} F = \oint [D\Psi^2 - Bg^{ij}\Psi_i\Psi_j + \tfrac{1}{2}k_c(\Delta_s\Psi)^2]dA. \quad (46a)$$

Choose orthonormal spherical harmonics with the conjugate-paired phase convention used for the amplitudes in the paper. They satisfy

$$\Delta_s Y_{lm} = -\frac{l(l+1)}{r_0^2} Y_{lm}. \quad (47)$$

Their orthogonality is with respect to $\sin\theta d\theta d\phi$; the surface area measure supplies an additional r_0^2 . Consequently

$$\delta^{(2)} F = \frac{k_c}{2} \sum_{l,m} \frac{|a_{lm}|^2}{r_0^2} [l(l+1) - 2] \left[l(l+1) - c_0 r_0 - \frac{\Delta p r_0^3}{2k_c} \right], \quad (48)$$

where

$$\Psi = \sum_{l,m} a_{lm} Y_{lm}, \quad l = 0, 1, \dots, \quad |m| \leq l, \quad a_{lm}^* = a_{l,-m}. \quad (49)$$

For clarity, the unfactored coefficient obtained from (45) is

$$\begin{aligned} r_0^2 D - l(l+1)B + \frac{k_c[l(l+1)]^2}{2r_0^2} &= \frac{k_c}{2r_0^2} \left\{ [l(l+1)]^2 \right. \\ &\quad \left. - \left[2 + c_0 r_0 + \frac{\Delta p r_0^3}{2k_c} \right] l(l+1) + 2c_0 r_0 + \frac{\Delta p r_0^3}{k_c} \right\}. \end{aligned} \quad (49a)$$

Factoring this polynomial gives (48). No assumption about m is needed.

Explicit elimination of the uniform mode. Before imposing a constraint, the first and second curvature-energy terms are

$$\begin{aligned} \delta F_c &= k_c c_0 r_0 (c_0 r_0 - 2) \frac{\sqrt{4\pi} a_{00}}{r_0} \\ &\quad + \frac{k_c}{2} \sum_{l,m} \frac{|a_{lm}|^2}{r_0^2} \left\{ [l(l+1)]^2 - (2 + 2c_0 r_0 - \tfrac{1}{2}c_0^2 r_0^2) l(l+1) + c_0^2 r_0^2 \right\}. \end{aligned} \quad (50)$$

The same substitution into (19) and (20) gives

$$\delta A = 2\sqrt{4\pi} a_{00} r_0 + \sum_{l,m} |a_{lm}|^2 [1 + \tfrac{1}{2}l(l+1)], \quad (51)$$

$$\delta V = \sqrt{4\pi} a_{00} r_0^2 + \sum_{l,m} |a_{lm}|^2 r_0. \quad (52)$$

For fixed area, a_{00} is a second-order correction determined by

$$\sqrt{4\pi} a_{00} = -\frac{1}{2r_0} \sum_{l,m} |a_{lm}|^2 [1 + \tfrac{1}{2}l(l+1)]. \quad (53)$$

The energy to compare is

$$\delta F = \delta F_c + \Delta p \delta V. \quad (54)$$

The substitution can be checked separately in its two parts:

$$\begin{aligned} \delta V &= \frac{r_0}{4} \sum_{l,m} |a_{lm}|^2 [2 - l(l+1)], \\ \delta F_c &= \frac{k_c}{2r_0^2} \sum_{l,m} |a_{lm}|^2 [l(l+1) - 2][l(l+1) - c_0 r_0]. \end{aligned} \quad (54a)$$

Adding them produces exactly (48). Terms containing the square of the induced a_{00} are fourth order and are omitted at this stage. For fixed volume instead,

$$\sqrt{4\pi} a_{00} = -\frac{1}{r_0} \sum_{l,m} |a_{lm}|^2, \quad (55)$$

$$\delta F = \delta F_c + \lambda \delta A = \delta F_c + \frac{1}{2} \left[\frac{k_c c_0}{r_0} (2 - c_0 r_0) - \Delta p r_0 \right] \delta A. \quad (56)$$

Now $\delta A = \frac{1}{2} \sum |a_{lm}|^2 [l(l+1) - 2]$, and (50) reduces to

$$\delta F_c = \frac{k_c}{2r_0^2} \sum_{l,m} |a_{lm}|^2 [l(l+1) - 2][l(l+1) - 2c_0 r_0 + \frac{1}{2} c_0^2 r_0^2]. \quad (56a)$$

The $\lambda \delta A$ term restores the second factor in (48). This is the paper's constraint check: the nonuniform quadratic mode energies agree, while the uniform amplitude is constrained rather than independently excited.

For $l \geq 2$, the first factor in (48) is positive. The second changes sign at

$$\Delta p_l = \frac{2k_c}{r_0^3} [l(l+1) - c_0 r_0]. \quad (57)$$

The $l = 1$ modes have zero quadratic energy because they translate the sphere. Among the nontrivial modes $l = 2$ reaches its threshold first. Its positive and negative axisymmetric amplitudes have the same quadratic energy, so their comparison requires the cubic calculation below.

4 Cylinder: equilibrium, modes, and instability

For a circular cylinder, $H = -1/(2\rho_0)$, $K = 0$, and $\Delta H = 0$. Equation (31) becomes

$$\Delta p \rho_0^3 + \lambda \rho_0^2 + \frac{1}{2} k_c (c_0^2 \rho_0^2 - 1) = 0. \quad (58)$$

The original long-cylinder approximation neglects the end caps, taking $L \gg \rho_0$. The energy is then

$$F = \Delta p \pi \rho_0^2 L + 2\pi \lambda \rho_0 L + \frac{1}{2} k_c (\rho_0^{-1} - c_0)^2 2\pi \rho_0 L. \quad (59)$$

The terminal $= 0$ printed in the paper's (59) applies once length equilibrium is imposed: this energy is proportional to L . The radius derivative gives (58), while the length derivative gives

$$\Delta p \rho_0^3 + 2\lambda \rho_0^2 + k_c (c_0 \rho_0 - 1)^2 = 0. \quad (60)$$

Subtracting twice (58) from (60) first yields $-\Delta p \rho_0^3 + 2k_c(1 - c_0 \rho_0) = 0$. Substituting back gives

$$\Delta p = 2k_c \rho_0^{-3}(1 - c_0 \rho_0), \quad (61)$$

$$\lambda = \frac{1}{2}k_c \rho_0^{-2}(3 - c_0 \rho_0)(c_0 \rho_0 - 1). \quad (62)$$

The physical axial coordinate used in the following metric is z , so write

$$\mathbf{Y} = (\rho_0 \cos \phi, \rho_0 \sin \phi, z), \quad 0 \leq \phi \leq 2\pi, \quad 0 \leq z \leq L. \quad (63)$$

In the printed (63), ρ_0 precedes the entire coordinate triple; its axial component must be read as above to agree with the paper's $g_{22} = 1$, axial interval, and Laplacian. Direct differentiation gives

$$\begin{aligned} g_{11} &= \rho_0^2, & g_{12} &= g_{21} = 0, & g_{22} &= 1, & g &= \rho_0^2, \\ L_{11} &= -\rho_0, & L_{12} &= L_{21} = L_{22} = 0, & \mathbf{n} &= (\cos \phi, \sin \phi, 0), & \Gamma_{ij}^k &= 0, \\ H &= -\frac{1}{2\rho_0}, & K &= 0, & KL^{11} &= KL^{12} = KL^{21} = 0, & KL^{22} &= -\rho_0^{-1}. \end{aligned} \quad (64)$$

The last row uses the cofactor interpretation of KL^{ij} specified after (2). It is essential not to set every KL^{ij} to zero merely because $K = 0$. The cylinder Laplacian and normal perturbation are

$$\Delta_c = \rho_0^{-2} \partial_\phi^2 + \partial_z^2, \quad (65)$$

$$\mathbf{Y}' = \mathbf{Y} + \Psi \mathbf{n}. \quad (66)$$

The periodic Fourier expansion is

$$\Psi = \sum_{m,n} b_{mn} e^{i[m\phi + 2\pi n z/L]}, \quad b_{mn}^* = b_{-m, -n}. \quad (67)$$

With $q = 2\pi \rho_0/L$, each mode has Laplacian eigenvalue $-(m^2 + n^2 q^2)/\rho_0^2$. Orthogonality therefore gives

$$\oint \Psi \Delta_c \Psi dA = -2\pi \rho_0 L \sum_{m,n} (m^2 + n^2 q^2) \frac{|b_{mn}|^2}{\rho_0^2}. \quad (68)$$

The companion integrals needed in (39) are

$$\begin{aligned} \oint \Psi^2 dA &= 2\pi \rho_0 L \sum |b_{mn}|^2, & \oint \Psi_\phi^2 dA &= 2\pi \rho_0 L \sum m^2 |b_{mn}|^2, \\ \oint \Psi_z^2 dA &= \frac{2\pi L}{\rho_0} \sum n^2 q^2 |b_{mn}|^2, & \oint (\Delta_c \Psi)^2 dA &= \frac{2\pi L}{\rho_0^3} \sum (m^2 + n^2 q^2)^2 |b_{mn}|^2. \end{aligned} \quad (68a)$$

Here and below sums without displayed indices still run over m, n . The area and volume expansions are

$$\delta A = \pi \rho_0 L \sum (m^2 + n^2 q^2) \frac{|b_{mn}|^2}{\rho_0^2} + 2\pi \rho_0 L \frac{b_{00}}{\rho_0} + O(b_{mn}^3), \quad (69)$$

$$\delta V = \pi \rho_0^2 L \sum \frac{|b_{mn}|^2}{\rho_0^2} + 2\pi \rho_0^2 L \frac{b_{00}}{\rho_0} + O(b_{mn}^3). \quad (70)$$

For example, (69) follows from $-2H\Psi = \Psi/\rho_0$ and $g^{ij}\Psi_i\Psi_j = \Psi_\phi^2/\rho_0^2 + \Psi_z^2$ in (19). Inserting the same integrals into the curvature part of (29) and (39) gives

$$\begin{aligned} \delta F_c = \frac{k_c\pi L}{\rho_0} & \left\{ (c_0^2\rho_0^2 - 1) \frac{b_{00}}{\rho_0} \right. \\ & + \sum \left[\left[\frac{1}{2}(c_0^2\rho_0^2 - 1) - 2c_0\rho_0 \right] (m^2 + n^2q^2) + 2c_0\rho_0 m^2 \right. \\ & \left. \left. + (m^2 + n^2q^2)^2 + 1 - 2m^2 \right] \frac{|b_{mn}|^2}{\rho_0^2} \right\} + O(b_{mn}^3). \end{aligned} \quad (71)$$

The total increment is

$$\delta F = \delta F_c + \Delta p \delta V + \lambda \delta A. \quad (72)$$

Using (61)–(62), the linear coefficient cancels through

$$(c_0^2\rho_0^2 - 1) + 4(1 - c_0\rho_0) + (3 - c_0\rho_0)(c_0\rho_0 - 1) = 0. \quad (72a)$$

In the quadratic coefficient, the pressure adds $2(1 - c_0\rho_0)$ and the tension adds $\frac{1}{2}(3 - c_0\rho_0)(c_0\rho_0 - 1)(m^2 + n^2q^2)$ to the bracket in (71). Collecting them gives

$$\delta F = \frac{k_c\pi L}{\rho_0} \sum [2c_0\rho_0(m^2 - 1) + (m^2 + n^2q^2)^2 - 4m^2 - 2n^2q^2 + 3] \frac{|b_{mn}|^2}{\rho_0^2}. \quad (73)$$

The area and volume check described in the paper. For a mode without a linear area or volume increment, set $b_{00} = 0$ at first order. Allow uniform second-order adjustments of ρ_0 and L . Equations (69)–(70) then impose

$$\begin{aligned} 2\pi L\delta\rho_0 + 2\pi\rho_0\delta L &= -\frac{\pi L}{\rho_0} \sum (m^2 + n^2q^2)|b_{mn}|^2, \\ 2\pi\rho_0 L\delta\rho_0 + \pi\rho_0^2\delta L &= -\pi L \sum |b_{mn}|^2. \end{aligned} \quad (73a)$$

Solving these two linear equations gives

$$\delta\rho_0 = \frac{1}{2\rho_0} \sum (m^2 + n^2q^2 - 2)|b_{mn}|^2, \quad \delta L = -\frac{L}{\rho_0^2} \sum (m^2 + n^2q^2 - 1)|b_{mn}|^2. \quad (73b)$$

At equilibrium the curvature-energy change of this uniform adjustment is $-\Delta p \delta V - \lambda \delta A$. Since it cancels the mode's area and volume increments, it adds precisely the pressure and tension terms of (72) to the mode's curvature energy. Thus the fixed-area, fixed-volume calculation also gives (73) for the nonuniform modes.

For rotational symmetry, put $m = 0$:

$$\delta F = \frac{k_c\pi L}{\rho_0} \sum_n [-2c_0\rho_0 + 2 + (n^2q^2 - 1)^2] \frac{|b_{0n}|^2}{\rho_0^2}. \quad (74)$$

The bracket can be nonpositive only if

$$c_0\rho_0 \geq 1. \quad (75)$$

The upper edge quoted in the paper is

$$n^2q^2 = \left(\frac{2\pi n\rho_0}{L} \right)^2 \leq 1 + \sqrt{2(c_0\rho_0 - 1)}. \quad (76)$$

Directly from (74), the complete condition also retains the lower edge:

$$|n^2 q^2 - 1| \leq \sqrt{2(c_0 \rho_0 - 1)}. \quad (76a)$$

Equation (76) alone is not sufficient when its corresponding lower bound is positive. For a sufficiently long cylinder the first marginal mode is

$$c_0 \rho_0 = 1, \quad n^2 q^2 = 1, \quad T = \frac{L}{n} = 2\pi \rho_0. \quad (77)$$

For $m = 1$ the same bracket reduces to $n^4 q^4$:

$$\delta F = \frac{k_c \pi L}{\rho_0} \sum_n n^4 q^4 \frac{|b_{1n}|^2}{\rho_0^2}. \quad (78)$$

Its $n = 0$ member translates the cylinder; $n \neq 0$ describes tube bending. For a real transverse deflection $a \cos(2\pi n z/L)$, the radial perturbation is $a \cos \phi \cos(2\pi n z/L)$. The four Fourier coefficients have magnitude $a/4$, so (73) gives $\delta F = k_c \pi \rho_0 L a^2 (2\pi n/L)^4/4$. Comparison with the quadratic bending energy of the tube centerline gives the modulus $k_c \pi \rho_0$ reported in the paper.

Completing the square in (73) gives

$$\delta F = \frac{k_c \pi L}{\rho_0} \sum [2(c_0 \rho_0 - 1)(m^2 - 1) + (m^2 + n^2 q^2 - 1)^2] \frac{|b_{mn}|^2}{\rho_0^2}. \quad (79)$$

At $n = 0$, its bracket factors as $(m^2 - 1)(m^2 - 3 + 2c_0 \rho_0)$. Thus the $m = 2$ cross-sectional mode becomes marginal at $c_0 \rho_0 = -1/2$, yielding the circular-to-elliptic instability discussed in the paper.

Recalculating Table I of the original paper. The first four data columns below are the original table's inputs. Its two radius estimates and curvature products follow from

$$\rho_0 = \frac{1}{2}(\rho_{\max} + \rho_{\min}), \quad c_0 \rho_0 = (c_0 \rho_{\max}) \frac{\rho_0}{\rho_{\max}}, \quad \rho_0 = \frac{T}{2\pi}. \quad (79a)$$

$\frac{1}{2}T$	$\rho_{\max}$	$\rho_{\min}$	$c_0 \rho_{\max}$	$\frac{1}{2}(\rho_{\max} + \rho_{\min})$	$c_0 \rho_0$	$T/(2\pi)$
1.8	0.6	0.5	1.2	0.55	1.100	0.573
1.4	0.6	0.3	1.4	0.45	1.050	0.446
1.2	0.6	0.2	1.6	0.40	1.067	0.382
1.0	0.6	0.1	1.8	0.35	1.050	0.318

For example, the second row gives $1.4(0.45/0.6) = 1.05$ and $2.8/(2\pi) = 0.4456\dots$. The original table prints 0.456 for this last entry; the recalculated value is shown above. These are the paper's data, not an additional chosen numerical example.

5 Cubic terms and the oblate–prolate comparison

Through third order, the geometric expansions in the paper are

$$\delta A = \oint [-2H\Psi + K\Psi^2 + \frac{1}{2}g^{ij}\Psi_i\Psi_j + (Hg^{ij} - KL^{ij})\Psi\Psi_i\Psi_j]dA, \quad (80)$$

$$\delta V = \oint (\Psi - H\Psi^2 + \frac{1}{3}K\Psi^3) dA. \quad (81)$$

The volume expression follows by integrating the parallel-area polynomial used in (20). For area, expand the square root of the determinant of (10) one order further. The cubic gradient term remains, while a constant normal offset has no cubic area term. Split the curvature energy by degree:

$$\delta F_c = \frac{1}{2}k_c\delta \oint (2H + c_0)^2 dA = \delta^{(1)}F_c + \delta^{(2)}F_c + \delta^{(3)}F_c. \quad (82)$$

Expanding the square gives the paper's decomposition

$$\begin{aligned} \delta^{(3)}F_c = 2k_c \left[\delta^{(3)} \oint H^2 dA + c_0 \delta^{(3)} \oint H dA \right] \\ + \frac{1}{2}k_c c_0^2 \oint (Hg^{ij} - KL^{ij})\Psi\Psi_i\Psi_j dA. \end{aligned} \quad (83)$$

The intermediate product rules needed to collect these terms are

$$\begin{aligned} \delta^{(3)} \oint H dA &= \oint [\delta^{(3)}H dA + \delta^{(2)}H \delta^{(1)}dA + \delta^{(1)}H \delta^{(2)}dA + H\delta^{(3)}dA], \\ \delta^{(3)} \oint H^2 dA &= \oint [2H\delta^{(3)}H dA + 2\delta^{(1)}H \delta^{(2)}H dA \\ &\quad + (2H\delta^{(2)}H + (\delta^{(1)}H)^2)\delta^{(1)}dA \\ &\quad + 2H\delta^{(1)}H \delta^{(2)}dA + H^2\delta^{(3)}dA]. \end{aligned} \quad (83a)$$

To carry out the underlying geometric expansion without introducing a new field, the exact normal of (5) can be written as

$$\mathbf{n}' = \frac{(1 - 2H\Psi + K\Psi^2)\mathbf{n} - g^{ij}[(1 - 2H\Psi)\Psi_i + \Psi L_{ik}g^{kl}\Psi_l]\mathbf{Y}_j}{\sqrt{g'/g}}. \quad (83b)$$

Equations (10), (83b), $L'_{ij} = \mathbf{Y}'_{ij} \cdot \mathbf{n}'$, and $2H' = g'^{ij}L'_{ij}$ specify a direct expansion to the required order. For example, since $\delta^{(3)}g_{ij} = 0$ in (10), the inverse relation gives

$$\delta^{(3)}g^{ij} = -g^{ik}(\delta^{(1)}g_{kl})(\delta^{(2)}g^{lj}) - g^{ik}(\delta^{(2)}g_{kl})(\delta^{(1)}g^{lj}). \quad (83c)$$

The corresponding curvature coefficient is

$$2\delta^{(3)}H = g^{ij}\delta^{(3)}L_{ij} + \delta^{(1)}g^{ij}\delta^{(2)}L_{ij} + \delta^{(2)}g^{ij}\delta^{(1)}L_{ij} + \delta^{(3)}g^{ij}L_{ij}. \quad (83d)$$

The paper prints the integrated component formulas as follows:

$$\begin{aligned} \delta^{(3)} \oint H dA &= \oint \left\{ \Psi^2 \nabla_i \Psi_j [(2H^2 - K)g^{ij} - HKL^{ij}] \right. \\ &\quad + \Psi\Psi_i\Psi_j [(4H^2 - \frac{3}{2}K)g^{ij} - 2HKL^{ij}] \\ &\quad + \frac{1}{2}\Psi_i\Psi_j \nabla_k \Psi_l [g^{-1}e_{3ik}e_{3jl} - g^{ij}g^{kl}] \\ &\quad + \frac{1}{2}\Psi^2\Psi_k \left\{ -Kg^{ij}\Gamma_{ij}^k + (2Hg^{ij} - KL^{ij}) \right. \\ &\quad \left. \left. \times [2(L_{jl}g^{lk})_i + K(g_{jl}L^{lk})_i + KL^{lk}g_{lm}\Gamma_{ij}^m] \right\} \right\} dA, \end{aligned} \quad (84)$$

$$\begin{aligned}
\delta^{(3)} \oint H^2 dA = \oint \bigg\{ & \Psi^3 (8H^5 - 12H^3 K + 4HK^2) \\
& + \Psi \Psi_i \Psi_j [(9H^3 - 4KH)g^{ij} + (K^2 - 5H^2 K)L^{ij}] \\
& + \Psi^2 \Psi_k \left\{ [(6H^2 - K)g^{ij} - 2HKL^{ij}](L_{jl}g^{lk})_i \right. \\
& \quad + (2KH^2 g^{ij} - HK^2 L^{ij})(g_{jl}L^{lk})_i \\
& \quad + [(K^2 - 2H^2 K)L^{ij} - KHg^{ij}]\Gamma_{ij}^k \\
& \quad \left. + [(4H^2 K - K^2)g^{ij} - HK^2 L^{ij}]L^{km}g_{ml}\Gamma_{ij}^l \right\} \\
& + \Psi^2 \Delta' \Psi (12H^3 - 7KH) + \Psi^2 \Delta'' \Psi (2K^2 H^{-1} - 6HK) \\
& + \frac{3}{2} H \Psi (\Delta' \Psi)^2 - \Psi (\Delta' \Psi) (\Delta'' \Psi) K H^{-1} \\
& + \frac{1}{2} \Psi_i \Psi_j \Delta' \Psi (Hg^{ij} + KL^{ij}) \\
& + \frac{1}{2} \Psi \Psi_k \Delta' \Psi \left\{ [(L_{jl}g^{lk})_i + KL^{km}g_{ml}\Gamma_{ij}^l]g^{ij} - KL^{ij}\Gamma_{ij}^k \right\} \\
& + \frac{H}{g} e_{3ik} e_{3jl} \Psi_k \Psi_l \nabla_i \Psi_j \bigg\} dA.
\end{aligned} \tag{85}$$

Here $\Delta' \Psi = g^{ij} \nabla_i \Psi_j$ and $\Delta'' \Psi = HL^{ij} \nabla_i \Psi_j$, as in the original paper. On a sphere both coincide with $\Delta_s \Psi$.

A discrepancy in the printed cubic formula. Equation (85) above is retained as printed. Under the sphere substitutions (42)–(44), its integral reduces to

$$\oint [H^3 \Psi g^{ij} \Psi_i \Psi_j + H^3 \Psi^2 \Delta_s \Psi + \frac{1}{2} H \Psi (\Delta_s \Psi)^2 + \frac{5}{2} H g^{ij} \Psi_i \Psi_j \Delta_s \Psi] dA. \tag{85a}$$

For $\Psi = r_0 a_2 P_2$, using $\Delta_s \Psi = -6\Psi/r_0^2$, this gives $48\pi a_2^3/7$. A direct expansion of the same surface instead gives $-48\pi a_2^3/35$ for the cubic term of $\oint H^2 dA$ and agrees with the paper's (89). Thus (85), as printed and with the stated definitions, is not verified by this spherical check. The following calculation derives (87)–(97) directly from the surface, without silently modifying (85) or assuming a corrected general formula.

Direct expansion for the paper's spherical deformation. The original axisymmetric perturbation is

$$\Psi = r_0 [a_0 + a_2 P_2(\theta)], \quad P_2(\theta) = \frac{1}{2} (3 \cos^2 \theta - 1). \tag{86}$$

Here a_0 and a_2 are dimensionless; a_0 will enforce fixed area. The following angular integrals supply all coefficients through cubic order:

$$\begin{aligned}
\int_0^\pi P_2 \sin \theta d\theta &= 0, & \int_0^\pi P_2^2 \sin \theta d\theta &= \frac{2}{5}, & \int_0^\pi P_2^3 \sin \theta d\theta &= \frac{4}{35}, \\
\int_0^\pi (\partial_\theta P_2)^2 \sin \theta d\theta &= \frac{12}{5}, & \int_0^\pi P_2 (\partial_\theta P_2)^2 \sin \theta d\theta &= \frac{12}{35}.
\end{aligned} \tag{86a}$$

They follow by inserting P_2 and integrating the polynomial in $\cos \theta$; for example $P_2^3 = (27 \cos^6 \theta - 27 \cos^4 \theta + 9 \cos^2 \theta - 1)/8$.

For a direct check using only the perturbation Ψ , the exact area element and principal curvatures of $\mathbf{Y}' = (r_0 + \Psi)\mathbf{n}$ are

$$dA' = (r_0 + \Psi) \sin \theta \sqrt{(r_0 + \Psi)^2 + \Psi_\theta^2} d\theta d\phi, \tag{86b}$$

$$c'_1 = \frac{(r_0 + \Psi)^2 + 2\Psi_\theta^2 - (r_0 + \Psi)\Psi_{\theta\theta}}{[(r_0 + \Psi)^2 + \Psi_\theta^2]^{3/2}}, \quad (86c)$$

$$c'_2 = \frac{r_0 + \Psi - \Psi_\theta \cot \theta}{(r_0 + \Psi)\sqrt{(r_0 + \Psi)^2 + \Psi_\theta^2}}, \quad H' = -\frac{1}{2}(c'_1 + c'_2).$$

These expressions result from differentiating the perturbed embedding in (86); they are intermediate geometry for this same calculation. Expansion of (86b) has no cubic area term on a sphere. The volume is the angular integral of $(r_0 + \Psi)^3/3$. Together with (86a) these give

$$\delta A = 4\pi r_0^2(2a_0 + a_0^2 + \frac{4}{5}a_2^2), \quad (87)$$

$$\delta V = 4\pi r_0^3(a_0 + a_0^2 + \frac{1}{5}a_2^2 + \frac{1}{3}a_0^3 + \frac{1}{5}a_0a_2^2 + \frac{2}{105}a_2^3). \quad (88)$$

For the curvature integrals, expanding (86b)–(86c) and using (86a) gives

$$\delta \oint H dA = -4\pi r_0 a_0 - \frac{12\pi r_0}{5}a_2^2 + \frac{12\pi r_0}{5}a_0a_2^2 - \frac{24\pi r_0}{35}a_2^3, \quad (88a)$$

$$\delta \oint H^2 dA = \frac{24\pi}{5}a_2^2 - \frac{48\pi}{5}a_0a_2^2 - \frac{48\pi}{35}a_2^3. \quad (88b)$$

In particular, a uniform radius change leaves $\oint H^2 dA = 4\pi$ unchanged; there are no pure a_0 terms in (88b). Combining $2k_c\delta \oint H^2 dA + 2k_c c_0\delta \oint H dA + \frac{1}{2}k_c c_0^2\delta A$ reproduces the paper's curvature-energy expansion:

$$\begin{aligned} \delta F_c = 4\pi k_c \Big[& (c_0^2 r_0^2 - 2c_0 r_0)a_0 + \frac{1}{2}c_0^2 r_0^2 a_0^2 \\ & + (\frac{12}{5} - \frac{6}{5}c_0 r_0 + \frac{2}{5}c_0^2 r_0^2)a_2^2 \\ & + (\frac{6}{5}c_0 r_0 - \frac{24}{5})a_0a_2^2 - (\frac{24}{35} + \frac{12}{35}c_0 r_0)a_2^3 \Big]. \end{aligned} \quad (89)$$

All expressions (87)–(89) retain total degree at most three in the amplitudes. Fixed area gives $a_0 = -2a_2^2/5 + O(a_2^4)$, so terms a_0^2 and $a_0a_2^2$ become fourth order or higher after the constraint is imposed. Equation (88) then reads

$$\eta = \frac{\delta V}{4\pi r_0^3/3} = -\frac{3}{5}a_2^2 + \frac{2}{35}a_2^3, \quad a_2^3 - \frac{21}{2}a_2^2 - \frac{35}{2}\eta = 0. \quad (90)$$

For small negative η , the two small roots are

$$\begin{aligned} a_2^+ &= 7[\cos(\theta_0 + \frac{4}{3}\pi) + \frac{1}{2}] > 0, \quad a_2^- = 7[\cos(\theta_0 + \frac{2}{3}\pi) + \frac{1}{2}] < 0, \\ \theta_0 &= \frac{1}{3} \arccos(1 + \frac{10}{49}\eta). \end{aligned} \quad (91)$$

To obtain them, substitute $a_2 = 7(\cos \theta_0 + 1/2)$ into the cubic: its left side before the η term is $343(\cos 3\theta_0 - 1)/4$. The other two branches give the small roots in (91). Their difference and sum are

$$a_2^+ - a_2^- = 7\sqrt{3} \sin \theta_0, \quad a_2^+ + a_2^- = 7(1 - \cos \theta_0). \quad (91a)$$

Multiplying them proves

$$(a_2^+)^2 - (a_2^-)^2 = 49\sqrt{3} \sin \theta_0 (1 - \cos \theta_0) > 0. \quad (92)$$

Fixed area reduces (89) to

$$\delta F_c = 4\pi k_c \left[\frac{2}{5}(6 - c_0 r_0) a_2^2 - \frac{12}{35}(c_0 r_0 + 2) a_2^3 \right]. \quad (93)$$

Using $a_2^3 = 21a_2^2/2 + 35\eta/2$ from (90) gives

$$\delta F_c = -4\pi k_c \left[\frac{1}{5}(24 + 20c_0 r_0) a_2^2 + 6(2 + c_0 r_0) \eta \right]. \quad (94)$$

At fixed η , the second term is identical for the two roots. Equation (92) then implies that the negative, oblate amplitude has lower energy when

$$c_0 r_0 < -\frac{6}{5} = -1.2. \quad (95)$$

Fixed pressure instead of fixed volume. At fixed area, add $\Delta p \delta V$ from (88) to (93):

$$\begin{aligned} \delta F = & \left[\frac{8\pi k_c}{5}(6 - c_0 r_0) - \frac{4\pi \Delta p r_0^3}{5} \right] a_2^2 \\ & + \left[-\frac{48\pi k_c}{35}(c_0 r_0 + 2) + \frac{8\pi \Delta p r_0^3}{105} \right] a_2^3. \end{aligned} \quad (95a)$$

Writing the excess above the $l = 2$ threshold as δp gives

$$\delta F = -\delta p \frac{4\pi r_0^3}{5} a_2^2 - \frac{32\pi k_c}{21} (c_0 r_0 + \frac{6}{5}) a_2^3 + \delta p \frac{8\pi r_0^3}{105} a_2^3, \quad (96)$$

$$\delta p = \Delta p - \Delta p_2 = \Delta p - 2k_c r_0^{-3} (6 - c_0 r_0). \quad (97)$$

Indeed, the quadratic coefficient in (95a) cancels at $\Delta p = \Delta p_2$. For the cubic coefficient at that pressure,

$$-\frac{48}{35}(c_0 r_0 + 2) + \frac{16}{105}(6 - c_0 r_0) = -\frac{32}{21}(c_0 r_0 + \frac{6}{5}). \quad (97a)$$

Thus the same $-6/5$ criterion is obtained at the onset in the fixed-pressure calculation. As the original paper states, a cubic expansion determines this comparison near threshold but does not determine the finite deformation amplitude: the fourth-order term has not been calculated there. This is the endpoint of the reproduction.

References

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- [1] Ou-Yang Zhong-can and Wolfgang Helfrich, “Bending energy of vesicle membranes: General expressions for the first, second, and third variation of the shape energy and applications to spheres and cylinders,” *Physical Review A* **39**, 5280–5288 (1989). [doi:10.1103/PhysRevA.39.5280](https://doi.org/10.1103/PhysRevA.39.5280).